Economics 5040: Dynamic Stochastic Modeling
Lecture 1: Economies, Preferences, and Intertemporal Prices

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1 What Is an Economy?

Before we study dynamic economies, we need to be precise about what economists mean by an *economy*.

An economy consists of six things:

1. **A list of agents**

   These are the people or households who make economic decisions.

2. **A specification of the preferences of each agent**

   Preferences describe how an agent ranks alternative consumption plans.

3. **A specification of the endowment of each agent**

   An endowment describes the goods or resources that an agent initially owns.

4. **A list of firms**

   Firms are the productive units in the economy.

5. **A specification of the production possibilities of each firm**

   This describes the ways in which a firm can transform inputs into outputs.

6. **A specification of the ownership of the profits of each firm**

   Ultimately, the profits of firms accrue to agents. We therefore need to know which agents own each firm and in what proportions.

These six ingredients describe the economic environment. Once they have been specified, we can ask how agents behave and how their separate decisions fit together.

Our behavior assumptions are these. An agent takes prices as given and maximizes his well-being subject to a budget constraint. A firm take prices as given and maximizes profits, given its production set.

2 Exchange Economies

An *exchange economy* is a particularly simple economy in which there is no production.

There are therefore no firms and no production technologies. Agents begin with endowments of goods and may trade those goods with one another.

An exchange economy is useful because it allows us to study preferences, prices, individual choice, and equilibrium without yet having to study production. Then the agent's wealth is just the value of her endowment.

We will begin even more simply: with one person choosing between two commodities.

3 Preferences and Utility

Suppose that x and y are two possible consumption bundles. We write

$$x \succeq y$$

to mean that the agent regards x as at least as good as y.

The preference relation $\succeq$ is the primitive economic object. It tells us how the individual ranks alternatives.

A *utility function* is a convenient numerical representation of preferences. A function u represents the preference relation if

$$x \succeq y \quad \Longleftrightarrow \quad u(x) \geq u(y).$$

Utility therefore should not be interpreted as a directly measurable quantity of happiness. Its purpose is to preserve the ranking of alternatives.

If u represents a set of preferences and g is any strictly increasing function, then

$$v(x) = g(u(x))$$

represents exactly the same preferences.

Thus the numerical values of utility themselves have no intrinsic meaning. What matters is their ordering.

Not every preference relation has a utility representation

Utility functions are extremely convenient, but not every possible preference relation can be represented by a real-valued utility function.

For example, lexicographic preferences over two goods give absolute priority to the first good: the consumer prefers any bundle with more of good 1, regardless of how much good 2 must be sacrificed. Only when two bundles have exactly the same amount of good 1 does the consumer compare their amounts of good 2.

Lexicographic preferences are complete and transitive, but they cannot be represented by an ordinary real-valued utility function. This example reminds us that utility representation requires more structure than rational ordering alone. In the models used in this course, we will normally assume preferences that do admit convenient utility representations.

4 Optimization with Two Commodities

Consider one consumer and two commodities. Let

$$x_1 \geq 0, \qquad x_2 \geq 0$$

denote the quantities consumed.

A consumption bundle is an ordered pair

$$x = (x_1, x_2).$$

Suppose preferences are represented by

$$u(x_1, x_2).$$

Let the prices of the two goods be $p_1 > 0$ and $p_2 > 0$, and let the consumer have wealth m.

The consumer's feasible choices satisfy

$$p_1 x_1 + p_2 x_2 \leq m.$$

The consumer therefore solves

$$\max_{x_1, x_2 \geq 0} u(x_1, x_2)$$

subject to

$$p_1 x_1 + p_2 x_2 \leq m.$$

This simple problem contains three distinct ideas:

- Preferences describe what the consumer wants.
- The budget constraint describes what the consumer can afford.
- Optimization selects the most preferred feasible bundle.

These three ingredients—preferences, feasibility, and optimization—will reappear throughout the course.

5 Relative Prices

If the consumer spends all available wealth, then

$$p_1 x_1 + p_2 x_2 = m.$$

Solving for x_2 gives

$$x_2 = \frac{m}{p_2} - \frac{p_1}{p_2} x_1.$$

The slope of the budget line is therefore

$$-\frac{p_1}{p_2}.$$

The ratio

$$\frac{p_1}{p_2}$$

is the *relative price* of good 1 in terms of good 2.

If the consumer gives up one unit of good 1, the market permits the consumer to obtain p_1/p_2 additional units of good 2.

Only relative prices matter for the consumer's tradeoffs. Multiplying all prices and wealth by the same positive number does not change the budget set.

6 Dated Commodities

The two commodities need not be physically different.

They can instead differ by the date at which they are delivered, their location, or the state of nature in which they are available.

Let c_0 be consumption today and c_1 be consumption next year. Even if the physical good is identical in both periods, c_0 and c_1 are different economic commodities because they are delivered at different dates.

Let p_0 be the price of one unit of consumption today and p_1 the price of one unit of consumption next year.

We can choose current consumption as the numéraire and set

$$p_0 = 1.$$

Suppose one unit of consumption today can be exchanged for R units of consumption next year. Then

$$R = 1 + r$$

is the *gross real interest rate*, where r is the net real interest rate.

The price today of one unit of consumption next year is therefore

$$p_1 = \frac{1}{R}.$$

Equivalently,

$$R = \frac{p_0}{p_1}.$$

Thus the gross real interest rate is a relative price:

$$\boxed{R = \text{the price of current consumption in units of future consumption.}}$$

One unit of consumption sacrificed today purchases R units of consumption next year. Conversely, one unit of next year's consumption has a current price of $1/R$.

This is our first example of an *intertemporal price*.

In the next lecture we will ask where this intertemporal tradeoff comes from when the economy can transform current output into future output through investment and production.

Figure 1 illustrates that

$$R = \frac{p_0}{p_1} = \frac{\partial u/\partial c_0}{\partial u/\partial c_1}.$$

This relationship is a necessary condition for utility maximization when the consumption choices are both strictly positive and the utility function is differentiable. The intuition is that the consumer is willing to cut back on current consumption–thus resulting in a high marginal utility of consumption now–when the rate of return from savings is high. Another way of thinking about this relative price is that a high real interest rate means that future consumption is cheap; so the consumer will give up consumption right now to get a lot of it in the future.

7 Time-Separable Preferences

In dynamic economic models, we often assume that preferences are *time separable*. In a two-period model, time-separable preferences can be represented in the form

$$U(c_0, c_1) = u(c_0) + \beta u(c_1),$$

where u is the period utility function and $\beta > 0$ determines the relative weight placed on future utility.

More generally, over $T + 1$ periods we might write

$$U(c_0, c_1, \ldots, c_T) = \sum_{t=0}^{T} \beta^t u(c_t).$$

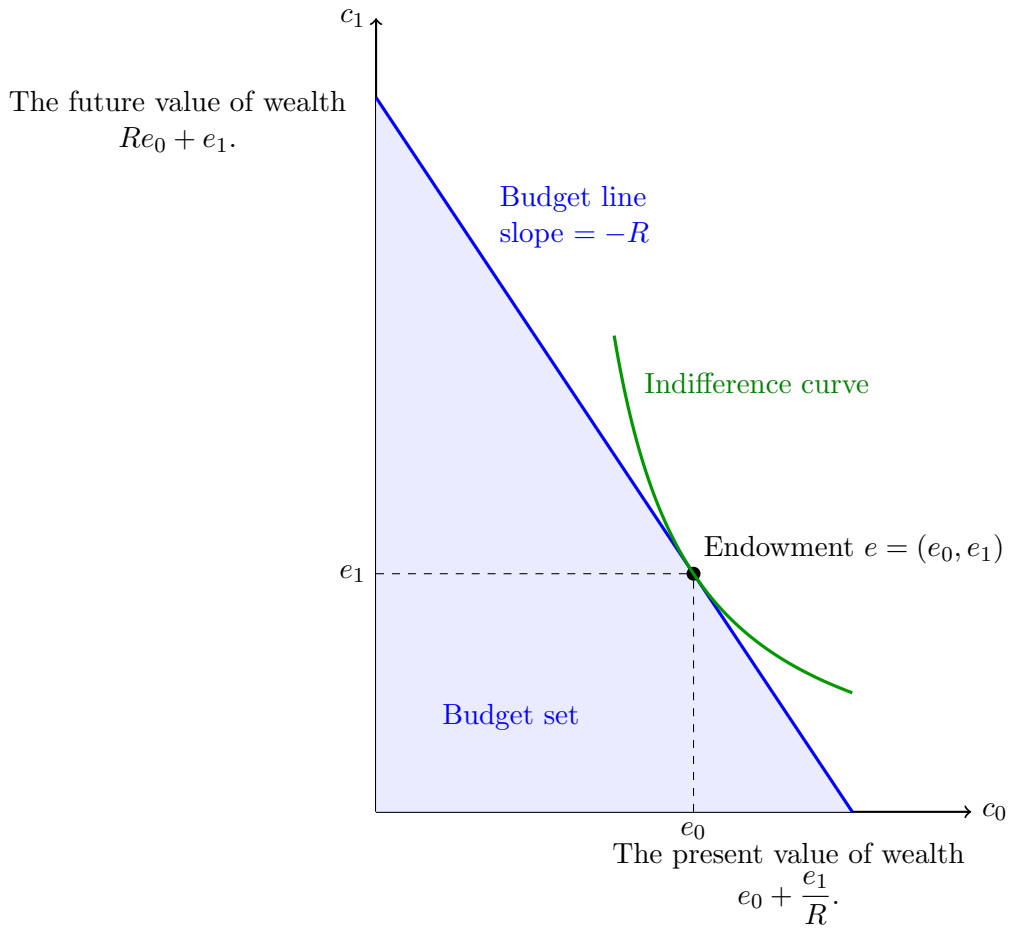


Figure 1: The gross real interest rate $R = 1 + r$ is the relative price of current consumption in terms of future consumption.

In "A Mathematical Theory of Saving " *Economic Journal*, vol. 38, no. 152, pp. 543–559, Frank Ramsey call $u(\cdot)$ *felicity*. We think of $U(c_0, c_1, \ldots, c_T)$ as the *utility* occurring from a lifetime plan of consumption. Felicity $u(c_0)$ is how happy you are today if you eat c_0 now. The functional form of your happiness relationship with food will not change tomorrow–you are always abstemious or you are always a glutton. You do, however, discount tomorrow's felicity by $0 \leq \beta \leq 1$.

Time separability is a very convenient assumption, but it is also a restrictive one. To see why, consider a much more familiar example. Suppose that the three commodities are breakfast, lunch, and dinner, and let c_0, c_1, and c_2 denote consumption at these three meals. Time-separable preferences would take the form

$$U(c_0, c_1, c_2) = u(c_0) + \beta u(c_1) + \beta^2 u(c_2).$$

Consider the consumer's subjective tradeoff between lunch and dinner. With time-separable preferences, this tradeoff depends on c_1 and c_2, but not on c_0. In particular, the marginal rate of substitution between lunch and dinner is

$$MRS_{1,2} = \frac{\beta u'(c_1)}{\beta^2 u'(c_2)} = \frac{u'(c_1)}{\beta u'(c_2)}.$$

Notice that breakfast, c_0, does not appear anywhere in this expression.

This implication is quite strong. In ordinary life, having eaten a large breakfast should certainly influence one's subjective tradeoff between lunch and dinner. After a very large breakfast, an additional unit of lunch might be considerably less attractive, perhaps making a person more willing to postpone consumption until dinner. Such an effect cannot occur with the time-separable specification above, because the marginal utility of lunch depends only on lunch consumption and the marginal utility of dinner depends only on dinner consumption. Time separability therefore rules out direct interactions between consumption at different dates.

We use time-separable preferences not because they provide a perfect description of human behavior, but because they make intertemporal choice especially tractable. They allow us to analyze the tradeoff between consumption at two dates without keeping track of the entire history of past consumption. Later in the course, this simplification will be particularly useful when we study dynamic optimization under uncertainty.